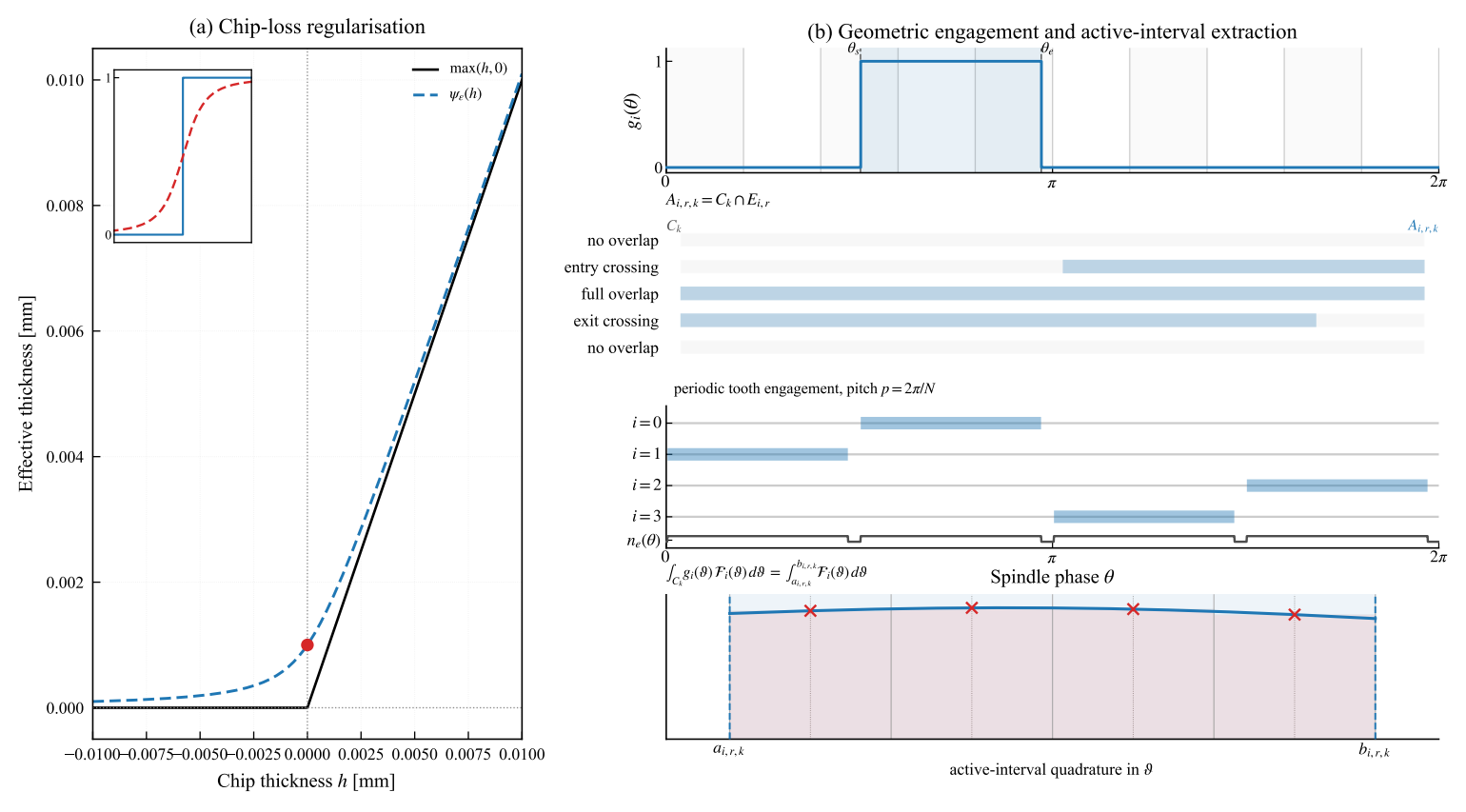

(a) Chip-loss regularisation
max(h, 0)
ψε(h)
Effective thickness [mm]
Chip thickness h [mm]
(b) Geometric engagement and active-interval extraction
gi(θ)
θs
θe
Ai,r,k = Ck ∩ Ei,r
Ck
Ai,r,k
no overlap
entry crossing
full overlap
exit crossing
no overlap
periodic tooth engagement, pitch p = 2π/N
i = 0
i = 1
i = 2
i = 3
ne(θ)
∫Ck gi(ϑ) Fi(ϑ) dϑ = ∫ai,r,k^bi,r,k Fi(ϑ) dϑ
Spindle phase θ
ai,r,k
bi,r,k
active-interval quadrature in ϑ